



Contrast-enhanced CT radiomics improves the prediction of abdominal aortic aneurysm progression

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Abstract

Objectives To determine if three-dimensional (3D) radiomic features of contrast-enhanced CT (CECT) images improve prediction of rapid abdominal aortic aneurysm (AAA) growth.

Methods This longitudinal cohort study retrospectively analyzed 195 consecutive patients (mean age, 72.4 years $\pm$ 9.1) with a baseline CECT and a subsequent CT or MR at least 6 months later. 3D radiomic features were measured for 3 regions of the AAA, viz. the vessel lumen only; the intraluminal thrombus (ILT) and aortic wall only; and the entire AAA sac (lumen, ILT, and wall). Multiple machine learning (ML) models to predict rapid growth, defined as the upper tercile of observed growth (> 0.25 cm/year), were developed using data from 60% of the patients. Diagnostic accuracy was evaluated using the area under the receiver operating characteristic curve (AUC) in the remaining 40% of patients.

Results The median AAA maximum diameter was 3.9 cm (interquartile range [IQR], 3.3–4.4 cm) at baseline and 4.4 cm (IQR, 3.7–5.4 cm) at the mean follow-up time of 3.2 ± 2.4 years (range, 0.5–9 years). A logistic regression model using 7 radiomic features of the ILT and wall had the highest AUC (0.83; 95% confidence interval [CI], 0.73–0.88) in the development cohort. In the independent test cohort, this model had a statistically significantly higher AUC than a model including maximum diameter, AAA volume, and relevant clinical factors (AUC = 0.78, 95% CI, 0.67–0.87 vs AUC = 0.69, 95% CI, 0.57–0.79; $p = 0.04$).

Conclusion A radiomics-based method focused on the ILT and wall improved prediction of rapid AAA growth from CECT imaging.

Key Points

- Radiomic analysis of 195 abdominal CECT revealed that an ML-based model that included textural features of intraluminal thrombus (if present) and aortic wall improved prediction of rapid AAA progression compared to maximum diameter.
- Predictive accuracy was higher when radiomic features were obtained from the thrombus and wall as opposed to the entire AAA sac (including lumen), or the lumen alone.
- Logistic regression of selected radiomic features yielded similar accuracy to predict rapid AAA progression as random forests or support vector machines.

Keywords Abdominal aortic aneurysm · Computed tomography · Machine learning · Thrombus

Abbreviations

3D Three-dimensional
AAA Abdominal aortic aneurysm

AUC Area under the receiver operating characteristic curve
CECT Contrast-enhanced CT
CI Confidence interval
CT Computed tomography
ICC Intraclass correlation coefficient
ILT Intraluminal thrombus

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IQR	Interquartile range
LR	Logistic regression
LSSPM	Level set shape prior method
ML	Machine learning
MR	Magnetic resonance
PCA	Principal component analysis
RBLow	Region between lumen and outer wall
RF	Random forest
RFE	Recursive feature elimination
RIL	Region within lumen
ROI	Region of interest
ROW	Region within outer wall
SVM	Support vector machine

Introduction

Abdominal aortic aneurysms (AAA) are common, with a pooled global prevalence of 4.8%, and are associated with significant morbidity and mortality [1, 2]. Clinical management of the 85–97% of AAA that are asymptomatic [3] is based on imaging features: those with maximum aneurysm diameter > 5–5.5 cm and/or growth rate > 1 cm/year are repaired. Repairs performed earlier than necessary confer no net mortality benefit [4]. The USA has one of the highest AAA repair rates in Western countries [5], with 30,000–40,000 repairs performed annually, most in intact AAA, at an annual cost of ~\$1B [6–8]. Nonetheless, AAA rupture is the 2nd most common non-traumatic vascular emergency in the USA [9].

Patients with AAA that do not meet repair criteria are followed with serial imaging at 6-month to 3-year intervals depending on their size [10, 11]. However, surveillance cannot predict future growth and a significant proportion of ruptured AAA—41% in a recent large single-center study [12]—have dimensions below repair thresholds at last surveillance. Accurate classification of future AAA growth in an individual could allow personalized stratification of surveillance intervals and better inform whether and when repair should be performed [13, 14].

Contrast-enhanced CT (CECT) is the reference standard to assess AAA morphology [10]. However, many factors believed to influence AAA progression, such as inflammation, hemodynamics, or biomechanics [15], are not readily accessible by CECT. Radiomics is a technique that extracts intensity and textural features of medical images that can offer information useful for phenotyping or staging various pathologies [16, 17], including atherosclerotic lesions seen in CECT [18–22]. The purpose of this study was to determine whether radiomics from CECT can predict those AAA that progress more rapidly in a large patient cohort with longitudinal imaging data.

Materials and methods

Study participants

The Institutional Review Board (IRB) at our institution approved this retrospective single-center study and waived the requirement for informed consent. Patients with a CECT depicting an AAA between January 2004 and December 2018 were included in this study based on the following criteria (Fig. 1): (a) AAA diameter ≥ 3 cm and (b) a follow-up CT (with or without contrast) or 3D MRI performed > 6 months after the baseline study. 3D MRI has been previously validated to accurately measure AAA maximum diameter compared to CECT, with a bias of < 1 mm, ICC > 0.99, and a coefficient of variation of < 2.8%, similar to the inherent inter-reader variability of CECT [23, 24]. Exclusion criteria were as follows: (a) aneurysm repair prior to or within the interval between the two studies, (b) AAA associated with dissection or occlusion, (c) incomplete anatomic coverage of the AAA in either exam, and (d) non-diagnostic image quality as determined by an abdominal radiologist with 6 years of experience (J.L.). Based on these criteria, 195 subjects were identified for inclusion in this study, of which 183 (93.8%) had CT as the follow-up study and 12 (6.2%) had 3D MRI as the follow-up study. The identified subjects were split with 60% ($n = 117$) used as the development cohort and 40% ($n = 78$) as the independent test cohort [25]. The development cohort was used to select the optimal radiomics model and to optimize two clinical predictive models (see “Statistical analysis”). The test cohort was used exclusively to compare the diagnostic accuracy of the fitted models.

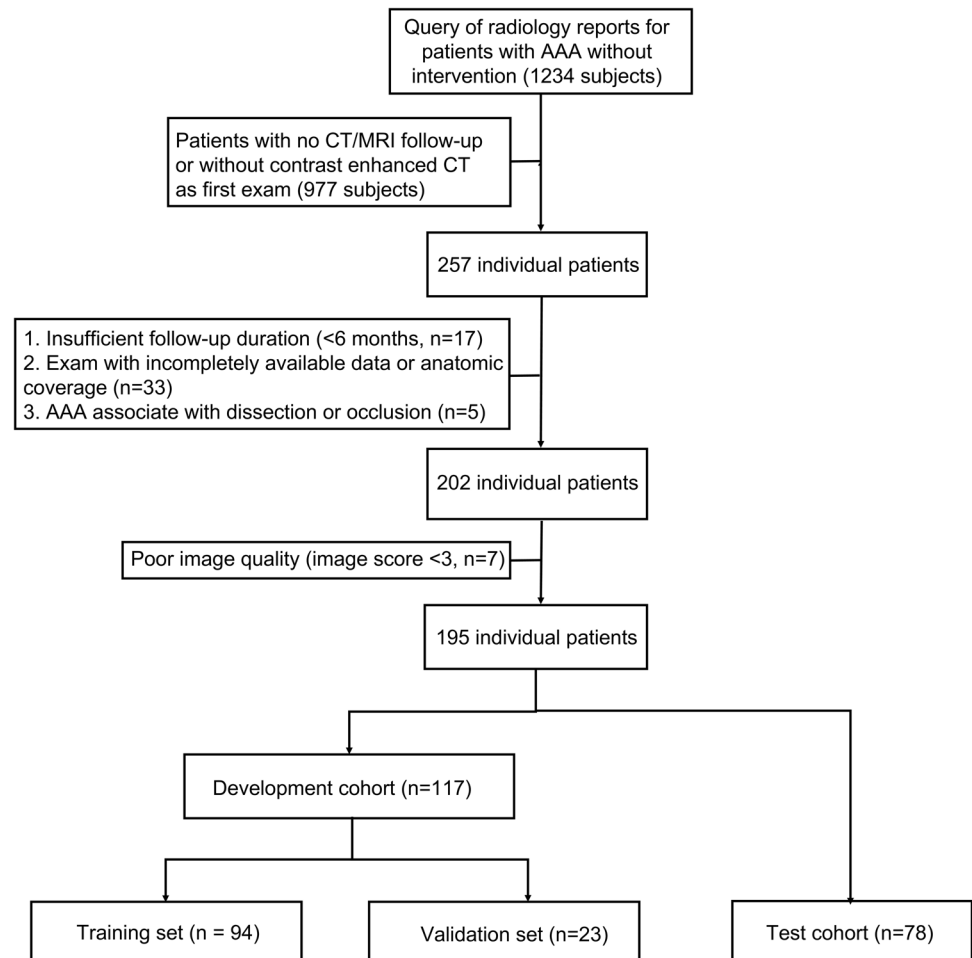
CT data acquisition

CT data were acquired according to institutional protocols using multidetector CT scanners (GE Healthcare and Philips Medical Systems platforms). All studies were acquired in the portal venous phase. A 120-kVp tube potential was common to all studies except those acquired using dual energy. Automatic tube current modulation was used for all exams. Images had varying slice thicknesses of 1.0–5.0 mm and in-plane resolutions of 0.7–1.0 mm.

Radiomics analysis

Radiomics analysis consisted of the following major steps: (a) AAA segmentation to define regions of interest (ROI), (b) image preprocessing, (c) radiomic feature extraction, and (d) radiomic feature selection and classification model fitting (Fig. 2).

Fig. 1 Workflow for inclusion and exclusion of study subjects



AAA segmentation and region-of-interest definitions

The AAA lumen and outer wall were first segmented using a previously validated semi-automatic level set shape prior segmentation method (LSSPM) [24]. Briefly, LSSPM adds an ellipse term in the classical level set equation [24, 26, 27]. To apply the method, an initial “seed” contour was manually delineated for the lumen and outer wall at one image slice by the operator. LSSPM then refined these contours and used them to seed contours in the next more cranial and caudal image slices, a process that was repeated automatically until all image slices were segmented. In cases where the segmentation outcome was deemed suboptimal by the operator, one or two additional manual seed contours were placed at the discretion of the operator to improve the overall segmentation result.

The lumen and outer wall segmentations were used to define three ROIs for radiomic analyses (Fig. 3). The first region included all voxels within the outer wall contour throughout the AAA extent (henceforth referred to as ROW), i.e., all image voxels containing the AAA lumen, thrombus (if present), and wall. The second ROI was defined

by the lumen contours (henceforth, RIL), thus containing only lumen voxels. Finally, the third ROI was defined as the region between the lumen and outer wall (henceforth, RBLOW), thus containing only the thrombus (if present) and wall tissue voxels.

Image preprocessing

All preprocessing was performed using built-in routines of the open-source PyRadiomics package (version 3.0; <https://pyradiomics.readthedocs.io/en/latest/>) [28], which was used throughout this work. CT images were first resampled to an isotropic voxel size of $1 \times 1 \times 1$ mm using B-spline interpolation (SimpleITK [29–31]). Normalization was subsequently applied by subtracting the mean value and dividing by the HU standard deviation. The dynamic range of the image was then scaled by 100 and limited to the range $[-300, 300]$. A fixed bin width of 25 HU was used for histogram and image discretization. In addition to the normalized CT images, filtered images were generated using Laplacian of Gaussian ($\sigma = 1.0, 2.0$, and 3.0) filters to detect fine to coarse textures, as well as 3D wavelet filters using a high (H) or a low (L)

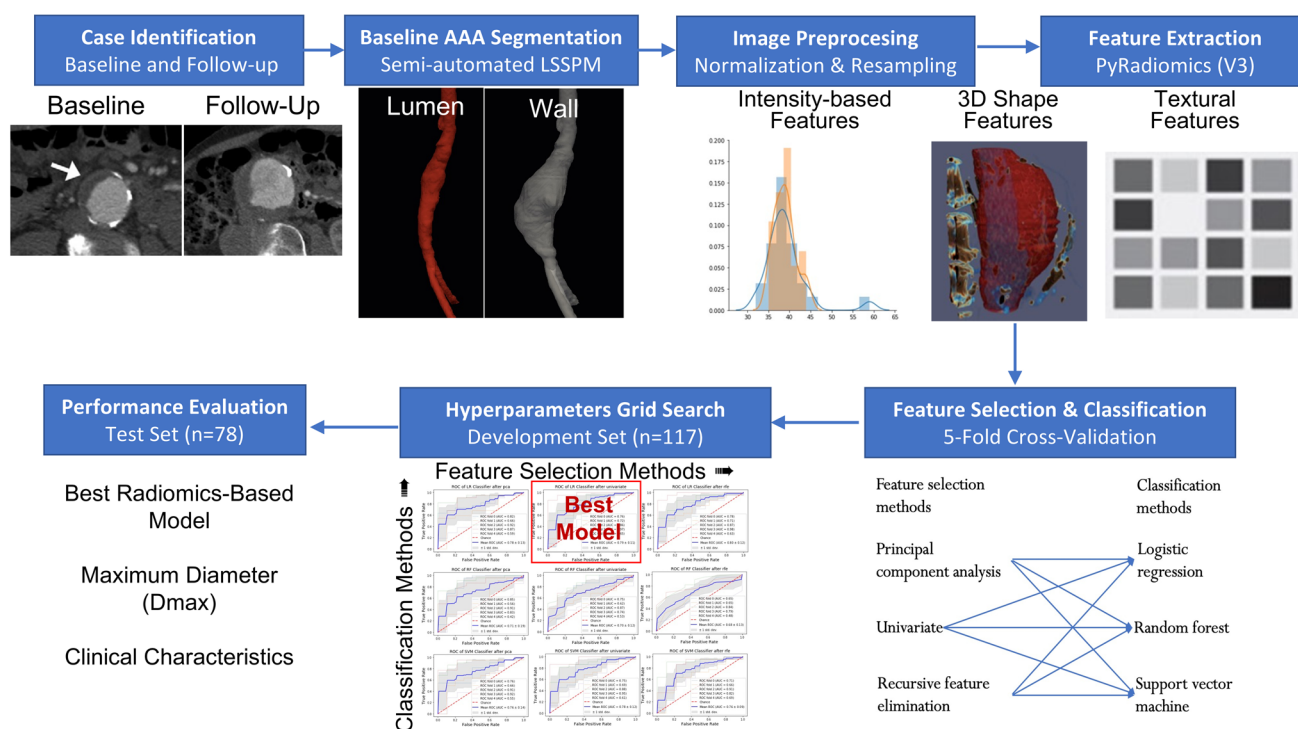


Fig. 2 Steps in radiomics analysis starting from subject screening, semi-automatic AAA segmentation to define region-of-interest (ROI), image preprocessing and feature extraction, development of ML models with feature selection and classification algorithms on the devel-

opment set using fivefold cross-validation, hyperparameters tuning based on Grid Search, and ROC performance comparison for the test set using the selected best radiomics-based ML model, maximum diameter ($D_{\max}$) and clinical characteristics

pass filter applied to each of the three spatial dimensions which yielded 8 decompositions (i.e., LLL, LLH, LHL, HLL, LHH, HLH, HHL, HHH, denoting the 3 spatial dimensions) to focus on textures of different spatial frequencies [22].

Radiomic feature extraction

The open-source PyRadiomics package (version 3.0; <https://pyradiomics.readthedocs.io/en/latest/>) [28] was used to extract 1130 features from the collection of images. Specifically for each ROI described above, 143 shape-based features were obtained, and from each filtered image, 19 image intensity first-order statistics and 74 texture-based features [32] of the image intensities within the ROI were also obtained. Z-score normalization was applied to each feature individually [33].

Feature selection and classification model development

Nine different ML models were tested, representing combinations of three feature selection methods and three classification methods. The feature selection methods were (a) univariate analysis, i.e., selecting features with the highest receiver operating characteristic (ROC) area under the curve (AUC) to predict the binary outcome (see below); (b)

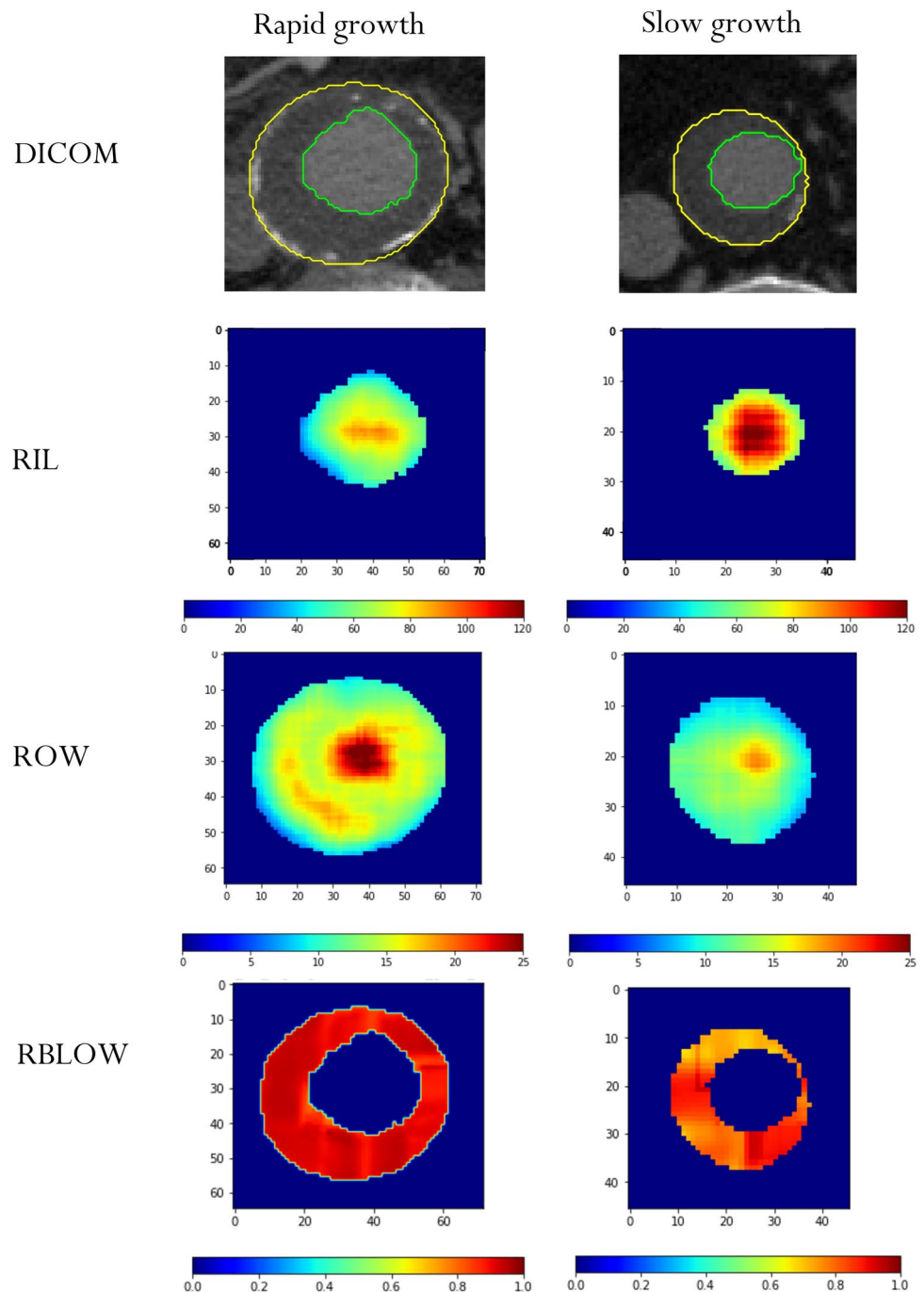
principal component analysis (PCA) [34, 35]; and (c) recursive feature elimination (RFE) [34, 35]. The three classification methods were (a) logistic regression (LR), (b) random forest (RF), and (c) support vector machine (SVM) [36, 37].

For each of the 9 models, a grid search was performed to optimize its hyperparameters in the development cohort using fivefold cross-validation (80% training, 20% validation). The hyperparameters yielding the highest average AUC on the 5 validation sets were selected for each model, and the best-performing model was selected for subsequent analyses.

Diameter and growth rate

Maximum AAA diameter was measured transverse to the vessel centerline using multi-planar reconstruction (MPR) in Horos (<http://www.horosproject.org>). An experienced radiologist and a vascular imaging scientist (B.T. and J.L., respectively) blinded to all other patient records [38] performed the diameter measurements in one half of the subjects each. Both readers initially measured the maximum diameter at baseline and follow-up for the same 15 subjects. The mean difference of those 30 measurements was 0.3 mm with a standard deviation of 1.6 mm, and an ICC of 0.99. Growth rate was calculated in millimeter/year as $[D_{\text{follow-up}} - D_{\text{baseline}} \text{ (mm)}] / (\text{follow-up time in years})$.

Fig. 3 Representative CECT images with segmentation contours for a fast-growing AAA (0.5 cm/year growth rate), and slow-growing AAA (0.01 cm/year). These contours were used to delineate the 3 separate regions from which radiomic features were extracted to predict fast-growing AAA (> 0.25 cm/year growth rate): the region within the lumen (RIL), the region within the outer wall (ROW), and the region between the lumen and outer wall (RBLW). The color maps below the CECT images show the voxel-wise value of the textural feature carrying the most information in each region of interest for predicting fast-growing AAA, as defined by the relative weights of features in the trained LR classifier. These were respectively: the original gray-level size-zone matrix feature non-uniformity for the ROW; the Laplacian of Gaussian $\sigma = 3.0$ -mm gray-level dependence matrix feature non-uniformity for the RIL; and the Laplacian of Gaussian $\sigma = 3.0$ mm 3D gray-level co-occurrence matrix feature maximal correlation coefficient for the RBLW. Detailed information and mathematical expressions for these radiomic features can be found at <https://pyradiomics.readthedocs.io/en/latest/features.html>



Statistical analysis

Statistical analysis was performed in GraphPad Prism 8.0 (GraphPad Software Inc.). Normality assumptions were assessed using the Kolmogorov–Smirnov test. Continuous and categorical variables were compared using the Mann–Whitney U test and χ^2 test, respectively. We compared the diagnostic accuracy to predict those patients that exhibited the upper tercile of growth rate in the test cohort of 3 models using the ROC

AUC and the DeLong method [39]. The three models were (1) the best-performing radiomics-based ML model selected from the development cohort as described above, (2) the maximum diameter, and (3) a LR model using maximum diameter, lumen, ILT, and total AAA sac volume [40, 41], as well as six relevant clinical characteristics (age, body mass index, smoking, diabetes, and hypertension status, as well as presence/absence of ILT [42]). The latter two models are referred to as clinical models below. The optimal parameters for the two

clinical models were also fitted using the development cohort using the identical cross-validation methodology as described above for the radiomics-based ML models. A two-sided p value less than 0.05 was considered statistically significant.

Results

Cohort characteristics

Patient characteristics are summarized in Table 1. Mean and median AAA diameter were 4.0 ± 0.8 and 3.9 cm (IQR: 3.3–4.4 cm) at baseline, and 4.6 ± 1.0 and 4.4 cm (IQR: 3.7–5.4 cm) at follow-up. The mean follow-up duration was 3.2 ± 2.4 years (range: 0.5–9 years), and the median was 2.6 years (IQR: 1.5–4 years). The median growth rate observed amongst all patients was 1.5 mm/year (IQR: 0.7–2.6 mm/year) and differed significantly ($p < 0.001$) between patients in the upper tercile (“rapid growth,” median: 3.5 mm/year, IQR: 2.9–4.5 mm/year) and the two lower terciles (“slow growth,” median: 1.1 mm/year, IQR: 0.6–1.6 mm/year). Finally, there were no differences in patient characteristics between development ($n = 117$) and test ($n = 78$) cohorts in age ($p = 0.88$), hypertension ($p = 0.89$), diabetes mellitus (DM; $p = 0.76$), smoking ($p = 0.94$), presence of ILT ($p = 0.70$), baseline diameter ($p = 0.87$), or growth rate ($p = 0.87$).

Optimal radiomics model

The radiomics model consisting of univariate selection of 7 features from the RBLOW region combined with LR classification achieved the highest AUC in the validation folds (AUC = 0.83, 95% confidence interval [CI]: 0.73–0.88, Fig. 4) and was selected as the “optimal” model for comparison to the clinical models. The selected features of the RBLOW region

used by the optimal model are shown in Fig. 5. Tables 2 and 3 provide the ROC AUCs in the development and test cohorts for features extracted from different regions using the optimal feature selection (univariate) and classification method (LR), as well as of all feature selection and classification methods using features extracted from the optimal region (RBLOW).

In the test cohort, the selected model had the 2nd highest accuracy (AUC = 0.78, 95% CI: 0.67–0.87). SVM instead of LR classification but also with univariate selection of RBLOW features achieved a slightly higher (but not statistically significantly so) AUC (AUC = 0.79, 95% CI: 0.68–0.87; Fig. 4).

Comparison of radiomics and clinical model

In the test cohort, the AUC of the selected radiomics-based ML model (AUC = 0.78, 95% CI: 0.67–0.87) was statistically significantly higher than that of either of the two clinical models. Specifically, the AUC of maximum diameter was 0.70 (95% CI: 0.58–0.79, $p = 0.04$, vs radiomics model) and that of maximum diameter plus clinical characteristics and AAA component volumes was 0.69 (95% CI: 0.57–0.79, $p = 0.04$, vs radiomics model; Fig. 6).

The accuracy of the radiomics-based ML model was not statistically significantly different when applied to only those subjects in the test set that had a baseline maximum AAA diameter of < 5.0 cm ($n = 70$, AUC = 0.76, 95% CI: 0.66–0.9, $p = 0.85$, vs accuracy when model applied to all subjects).

Discussion

AAA progression is an incompletely understood process, believed to be influenced by compositional features (e.g., presence of calcification [43] or ILT [42, 44, 45]), inflammation [46, 47] and biomechanical factors such

Table 1 Demographics and clinical characteristics of AAA patients

	All ($n = 195$)	Rapid growth ^a ($n = 57$)	Slow growth ^b ($n = 138$)	p value
Age (years), mean $\pm$ SD	72.4 $\pm$ 9.1	72.4 $\pm$ 9.1	72.3 $\pm$ 9.1	.99
Hypertension ^c , n , %	149 (76.4)	45 (78.9)	104 (75.4)	.71
DM ^d , n , %	37 (19.0)	7 (12.3)	30 (21.7)	.16
Smoking, n , %	57 (29.2)	21 (36.8)	36 (26.1)	.17
ILT	103 (52.8)	43 (75.4)	60 (43.5)	< .001
Baseline diameter (cm), median [IQR]	3.9 (3.3–4.4)	4.5 (3.9–5.2)	3.6 (3.3–4.1)	< .001
Follow-up diameter (cm), median [IQR]	4.4 (3.7–5.4)	5.5 (4.9, 5.9)	4.0 (3.5, 4.6)	< .001
Growth rate (mm/year), median [IQR]	1.5 (0.7–2.6)	3.5 (2.9–4.5)	1.1 (0.6–1.6)	< .001
Follow-up duration (years), mean $\pm$ SD	3.2 $\pm$ 2.4	2.6 $\pm$ 2.1	3.2 $\pm$ 2.3	.08

^aRapid growth defined as > 0.25 cm/year

^bSlow growth defined as ≤ 0.25 cm/year

^cHypertension defined as resting blood pressure of more than 140/90 mmHg

^dDiabetes mellitus

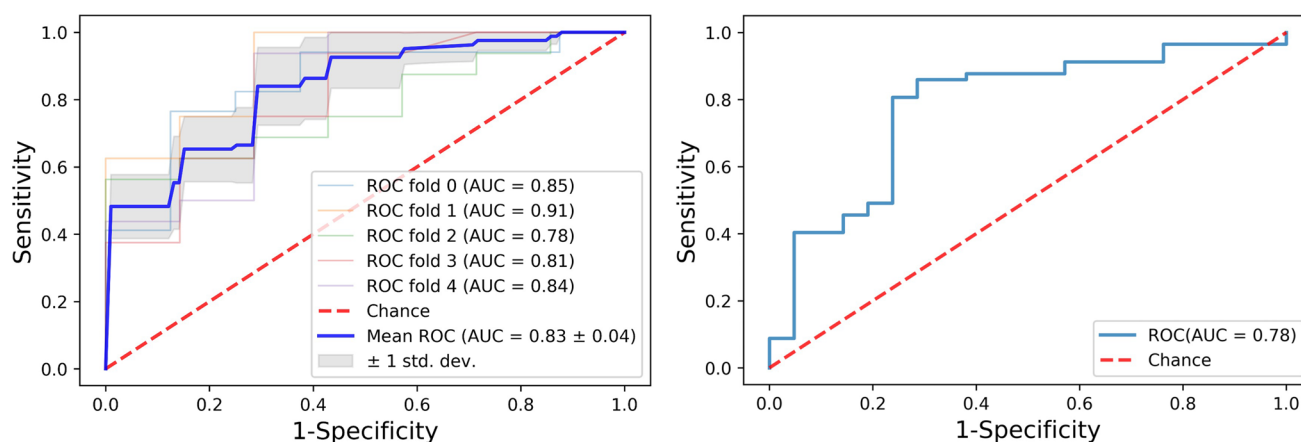


Fig. 4 Diagnostic accuracy in the development and test cohorts (left- and right-hand panels, respectively) to predict rapid AAA growth (>0.25 cm/year) for the optimal ML model identified (univariate selection of RBLOW features plus LR classifier; see text for details).

In the left-hand plot, shaded areas indicate the variance observed across the 5 cross-validation folds, and light-colored lines indicate the AUC achieved for each fold separately

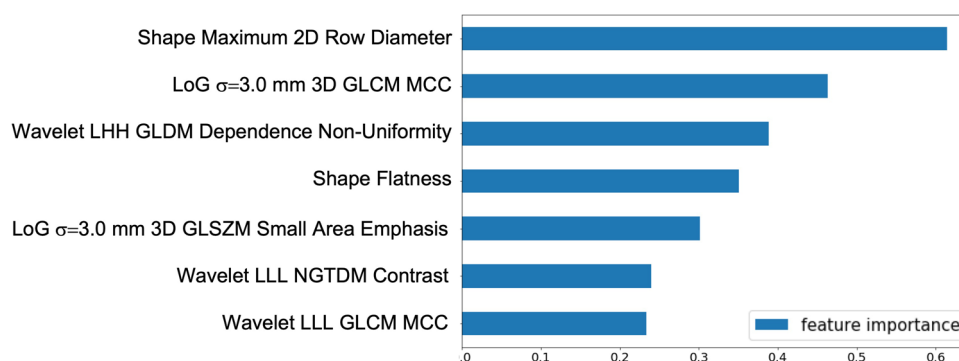


Fig. 5 Radiomic features selected in the optimal ML model to predict rapid AAA progression (>0.25 cm/year). LoG: Laplacian of Gaussian; GLCM: gray-level co-occurrence matrix; GLDM: gray-level dependence matrix; GLSZM: gray-level size zone matrix; NGTDM:

neighboring gray tone difference matrix; MCC: maximal correlation coefficient. Detailed information and mathematical expressions for these radiomic features can be found at <https://pyradiomics.readthedocs.io/en/latest/features.html>

Table 2 Discriminatory power to predict fast AAA growth (>0.25 cm/year) using different feature extraction regions for both the development and test cohorts. Results are shown only for the feature selection method (univariate) and classification algorithm (logistic regression; LR) that yielded the best-performing model overall

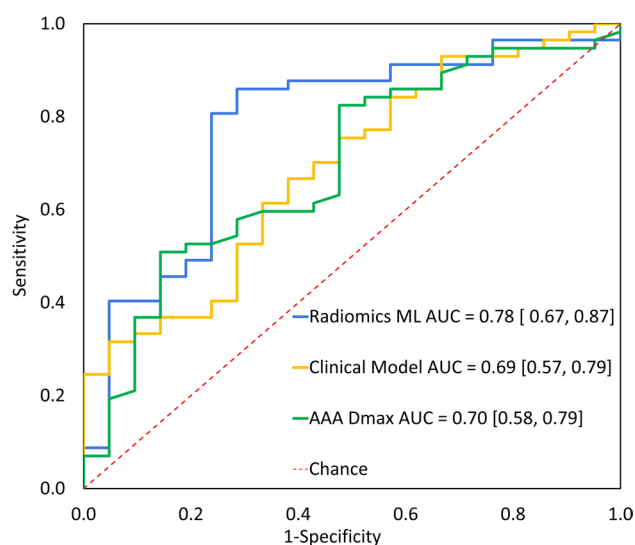
AUC [95% CI]	Selected model	Development cohort	Test cohort
RIL	Univariate + LR	0.73 [0.64, 0.81]	0.52 [0.41, 0.64]
ROW	Univariate + LR	0.81 [0.73, 0.88]	0.71 [0.59, 0.80]
RBLOW	Univariate + LR	0.83 [0.73, 0.88]	0.78 [0.67, 0.87]

as hemodynamics [40] and wall stress and strength [48]. Various markers using diverse imaging modalities including MRI, CECT, and 3D ultrasound have been developed that can each assess a subset of these factors for the purpose of predicting progression [49]; however, maximum

diameter remains the only validated independent prognostic marker [13]. The present work suggests that radiomics of the AAA thrombus and wall tissue from a single CECT scan can predict the 1/3rd of AAA that will exhibit the fastest growth rate in the following 3 years. A tool that can predict future growth of an individual AAA can potentially improve both the outcomes and costs associated with AAA. International guidelines recommend follow-up every 3 years for aneurysms 3 to 3.9 cm in diameter, annually for aneurysms 4 to 4.9 cm, and every 3 to 6 months for aneurysms ≥ 5 cm, with the purpose of detecting when guideline-recommended repair thresholds based on diameter and growth rate are reached [10]. Periodic surveillance is well suited for timing repair for the majority of AAA [13, 50, 51] whose growth rate is 0.2–0.3 cm/year [52, 53]. However, AAA at the highest risk of progressing to rupture exhibit a faster growth rate [53, 54]. Surveillance

Table 3 Discriminatory power of radiomics-based ML models on the (a) development cohort and (b) test cohort to predict AAA growth using RBLOW features

AUC [95% CI]	PCA	Univariate	Ref
(a) Development cohort			
LR	0.73 [0.63, 0.80]	0.83 [0.73, 0.88]	0.82 [0.73, 0.88]
RF	0.68 [0.58, 0.76]	0.79 [0.70, 0.86]	0.75 [0.67, 0.83]
SVM	0.76 [0.65, 0.82]	0.78 [0.69, 0.85]	0.82 [0.73, 0.88]
(b) Test cohort			
LR	0.63 [0.52, 0.74]	0.78 [0.67, 0.87]	0.73 [0.62, 0.82]
RF	0.71 [0.60, 0.80]	0.78 [0.68, 0.87]	0.77 [0.66, 0.86]
SVM	0.75 [0.64, 0.84]	0.79 [0.68, 0.87]	0.73 [0.62, 0.83]

**Fig. 6** Comparison of ROC performance for the test set between the radiomics-based ML model with 7 features extracted from RBLOW, the maximum diameter (Dmax), and clinical characteristics-based models

can neither be closely adhered to for all patients [55, 56], nor can it predict future rapid growth. Furthermore, the cost of preventing a single AAA rupture-related death is estimated at ~\$1 M [57], and the timing of intervention is a major driver of this cost. A tool for classifying an individual AAA into rapid or slow growth could help optimize surveillance intervals and improve the decision of when to intervene.

At a basic level, we hypothesized that a radiomic analysis of AAA extracted from abdominal CECT may be able to establish one or more factors that affect AAA progression via their influence on tissue radiopacity. We observed that 7

radiomic features of the ILT and aortic wall, selected using a development cohort ($n = 117$), improved prediction of AAA that grew more rapidly compared to maximum diameter (AUC = 0.78 versus 0.70, Fig. 6) in a relatively large held-out test cohort ($n = 78$). The selected features included two shape-based radiomic features, one related to AAA maximum diameter (shape maximum 2D row diameter, Fig. 5), but also 5 radiomic features related to image intensity and texture (e.g., gray-level co-occurrence matrix eigenvalues, Figs. 3, 5).

Radiomic analyses of intact unrepaired AAA CECT have not been previously reported, and therefore, there is no literature base against which our findings can be compared. A prior study reported that a sac texture at CECT after endovascular repair predicts progression [58]. In that setting, endoleak, a major risk factor for progression [59], is reasonably expected to alter the sac's textural image features. In this study, given the absence of such a direct link, we explored if focusing radiomic features to specific anatomic regions would alter prediction accuracy. For example, radiomic features of the lumen (RIL region) may reflect primarily hemodynamic conditions, since contrast is carried by flowing blood. Inhomogeneous intraluminal contrast opacification, likely arising from slow or recirculating flow, has been previously associated with rapid AAA progression [29]. Similarly, radiomic features of the ILT and wall (RBLOW region) may reflect increased pro-inflammatory neovasculation [30]. Contrast enhancement of the AAA wall differs amongst patients [31] and delayed enhancement has been found to correlate with serum levels of C-reactive protein [60]. In each scenario, radiomic features of the corresponding region are expected to reflect the related processes.

We found that radiomics of the ILT and wall were more accurate in predicting AAA progression compared to radiomics of the lumen, or of the combined lumen, ILT and wall regions. Thus, radiomics offers information that complements the analysis of direct observables such as ILT burden (presence, thickness, volume, etc.), which we have previously shown to correlate with AAA progression [42]. However, this finding should be interpreted with caution. One limitation, for example, is that shape radiomic features calculated from the lumen region only are unlikely to reflect AAA dimensions, as the majority of AAA have a relatively small-caliber lumen (with adjacent ILT accounting for overall dimensions). In contrast, shape features of the other two regions we explored (ROW and RBLOW) include a feature akin to the transverse AAA diameter that is often used clinically [10]. Interestingly, once the radiomic feature extraction region and feature selection method were fixed, we observed only small differences in diagnostic accuracy using different classification models. Tested models included random forests and support vector machines which have degraded performance with limited training data. The limited differences

in predictive accuracy between classification methods offers some confidence for our findings. Conversely, the feature selection method had a relatively larger impact on model accuracy, with the univariate method yielding the highest predictive accuracy in nearly all cases. This agrees with previous reports [61].

There are several limitations to our study that preclude our results from being generalized. We did not include any female patients, given the demographics of the population served at our institution. Nonetheless, AAA are approximately 3–6 times more common in men than women [15] so that our results apply to a substantial proportion of the affected population. Furthermore, radiomic features can be affected by variations in scanner vendors imaging protocols and image reconstruction algorithms, as well as segmentation algorithms and potentially patient characteristics. Although many of these factors varied significantly in our data set given the 14-year period covered, there was insufficient data to assess the impact of individual variables on either the final model or on individual radiomic feature reproducibility. Their impact should be carefully characterized in further research [62], and our results should ideally be confirmed on additional external data sets that control for these variables to ensure generalizability. Finally, test characteristics reported in this work can be used to guide selection of sample sizes needed in future studies to ensure adequate power to detect a significant increase in diagnostic accuracy for predicting fast-growing AAA.

In summary, this work suggests that radiomic analysis of the AAA intraluminal thrombus and wall in abdominal CECT studies routinely acquired for surveillance may improve detection of those AAA that are likely to progress more rapidly. If the technique proves to be robust in future prospective studies, it could offer a much-needed clinical tool to assist in deciding when and whether to repair a AAA.

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Declarations

Guarantor The scientific guarantor of this publication is Dimitrios Mitsouras.

Conflict of interest Fei Xiong is currently an employee of Siemens Medical Solutions USA, Inc. This research work was completed during her graduate study at USCF; there is no relevance to her current role.

Statistics and biometry One of the authors has significant statistical expertise.

Informed consent Written informed consent was waived by the Institutional Review Board.

Ethical approval Institutional Review Board approval was obtained.

Study subjects or cohorts overlap Some study subjects or cohorts have been previously reported in

Zhu C, Leach JR, Wang Y, Gasper W, Saloner D, Hope MD. Intraluminal thrombus predicts rapid growth of abdominal aortic aneurysms. *Radiology*. 2020;294(3):707–13.

Methodology

- retrospective
- experimental
- performed at one institution

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